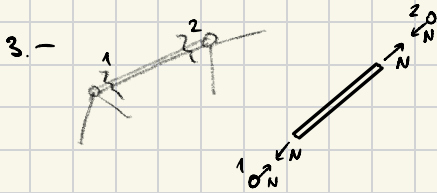


ESTRUCTURAS ARTICULADAS

1.- Todo los nudos son articulados

2.- Barras sometidas a AXIL.

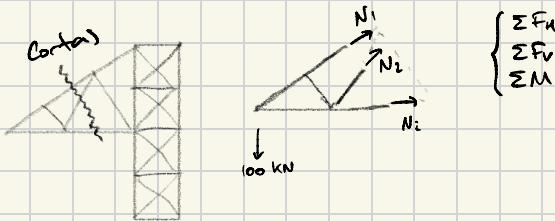


4.- Resolución:

a) Método de los nudos: EC: $\begin{cases} \sum F_H = 0 \\ \sum F_V = 0 \end{cases}$

Como mucho puedes gestionar 2 incógnitas (2 ecuaciones) $\rightarrow$

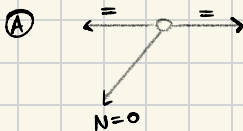
b) Método de las secciones:



Contas por tres barras y no hace falta resolver ningún nudo.

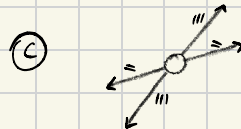
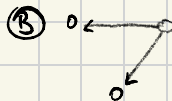
$\oplus$ MUY ÚTIL

5.- Nudos especiales.



Tres barras y dos a continuación

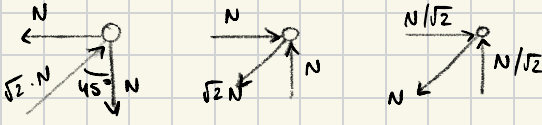
(NUDO EN T)



Deben ser colineales.

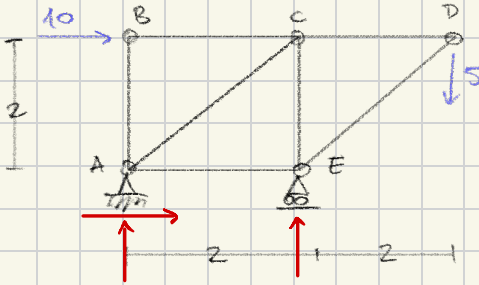
(NUDO EN CRUZ)

©



(NUDO A 45°)

6.- Ejemplo:



FÓRMULA DE GH.

Ligaduras externas
(reacciones)

Nº nudos

$$GH = LE + 6 - 2N$$

Nº de barras

$$GH = 3 + 7 - 2 \cdot 5 = 0 \text{ ISOSTÁTICA}$$

¿CORRECTO?

$$GH = GH1 + GHE$$

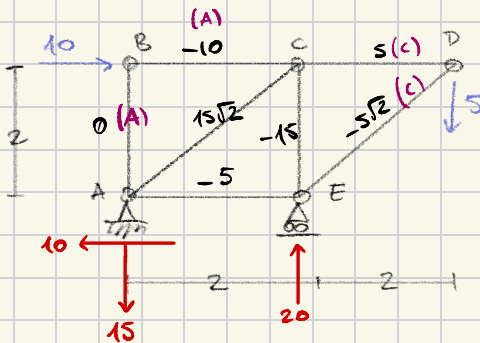
$$GHE = \text{Reacc.} - 3 \text{ ecuac de equl}$$

REACCIONES:

$$\sum M_A = 0 \rightarrow 2R_E = 4 \cdot 5 + 2 \cdot 10 \rightarrow R_E = 20 \text{ KN}$$

$$\sum F_v^{\uparrow} = 0 \rightarrow R_A + R_E - 5 = 0 \rightarrow R_A = -15 \text{ KN}$$

$$\sum F_H = 0 \rightarrow H_A + 10 = 0 \rightarrow H_A = -10 \text{ KN}$$



No es nudo especial.

© } Bueno, en realidad es un nudo en 45°.

Pero... se hacen así:

$$\sum F_H = 0 \rightarrow 10 + 5 - \frac{N_{CA}}{\sqrt{2}} = 0$$

$$N_{CA} = 15\sqrt{2}$$

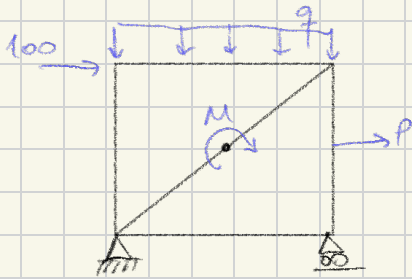
$$\sum F_v = 0 \rightarrow N_{CE} - \frac{N_{CA}}{\sqrt{2}} = 0$$

$$N_{CE} = -15$$

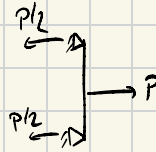
$$N_{AE} - 10 + \frac{N_{AC}}{\sqrt{2}} = 0$$

$$N_{AE} = -5$$

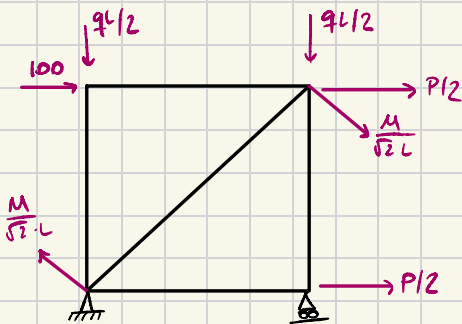
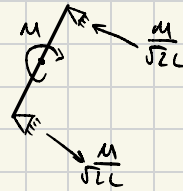
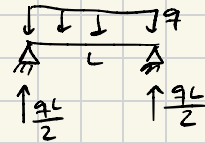
7.- Cargas en el interior de barras.



Transformación



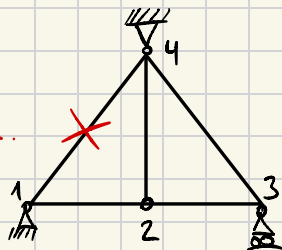
La teoría:



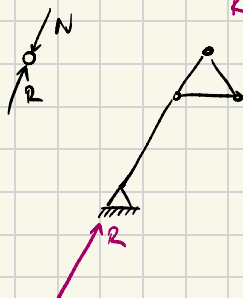
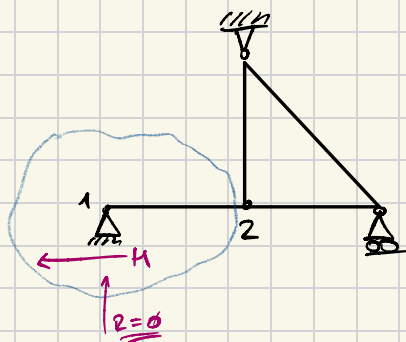
TEOREMAS

1.- Barra entre apoyos fijos

Se elimina.



2.- Barra solitaria con apoyo extremo



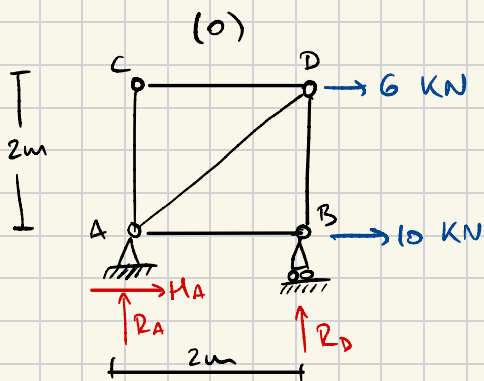
Reacción colineal con la barra

NOTA:

Si no lo piden explícitamente, no calcular las reacciones

3. Teorema de Castigliano

⊕ Ejemplo: Calcular $\vec{u}_B$ y $\vec{u}_D$



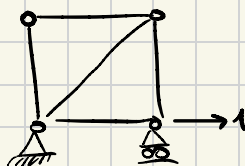
1) G_H : $G_H = LE + b - 2 \cdot n$

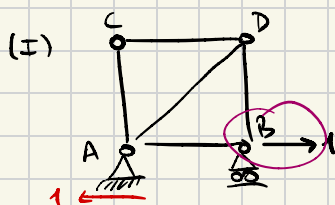
$G_H = 3 + 5 - 2 \cdot 4 = 0$. Isostática.

2. $\vec{u}_B$ y $\vec{u}_D$

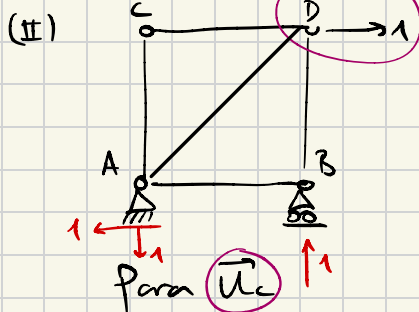
Utilizamos un esquema auxiliar.

(I)





Para $\vec{u}_B$



Para $\vec{u}_C$

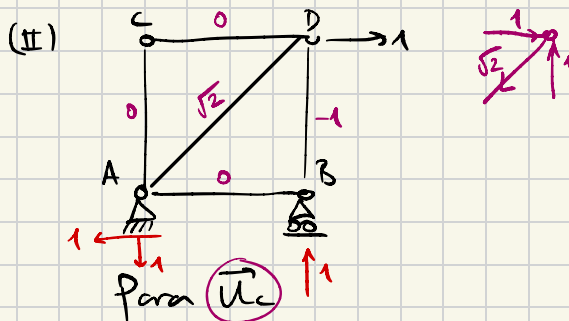
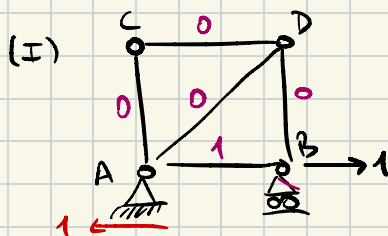
Esquemas 0, I, II

$$N_i^{(0)} = 10 \cdot N_i^I + 6 N_i^{II}$$

$$\vec{M}_B = \sum \frac{N_i^{(0)} \cdot N_i^{(I)} \cdot L_i}{EA_i} = \frac{1}{EA} \sum N_i^0 \cdot N_i^I \cdot L_i =$$

$$= \frac{1}{EA} \left[10 \sum N_i^{I^2} \cdot L_i + 6 \sum N_i^{II} \cdot N_i^I \cdot L_i \right]$$

$$\vec{u}_D = \frac{1}{EA} \sum N_i^0 \cdot N_i^{II} \cdot L_i = \frac{1}{EA} \left[10 \sum N_i^I \cdot N_i^{II} \cdot L_i + 6 \sum N_i^{II^2} \cdot L_i \right]$$



Creamos una tabla:

Barra	N_i^I	N_i^II	L_i	$N_i^I N_i^{II} L_i$	$N_i^{I^2} L_i$	$N_i^{II^2} L_i$	N_i^0
AB	1	0	2	0	2	0	10
AC	0	0	2	0	0	0	0
AD	0	$\sqrt{2}$	$2\sqrt{2}$	0	0	$4\sqrt{2}$	$6\sqrt{2}$
CD	0	0	2	0	0	0	0
BD	0	1	2	0	0	2	-6
				0	2	$2+4\sqrt{2}$	

$$\vec{M}_B = \sum \frac{N_i^{(0)} \cdot N_i^{(I)} \cdot L_i}{EA_i} = \frac{1}{EA} \sum N_i^0 \cdot N_i^I \cdot L_i = \frac{1}{EA} \left[10 \sum N_i^{I^2} L_i + 6 \sum N_i^{II} N_i^I L_i \right]$$

$$\vec{M}_B = \frac{10 \cdot 2 + 6 \cdot 0}{EA}$$

También puedes:

$$\Delta L = \frac{NL}{EA}$$

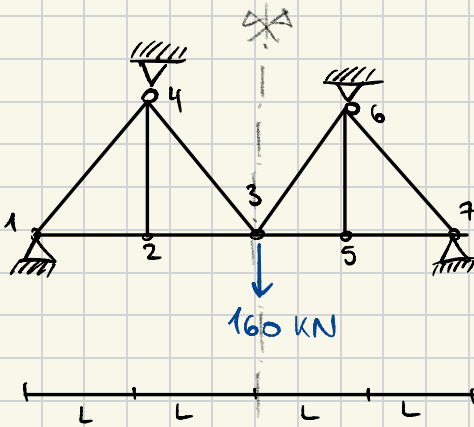


$$\vec{M}_D = \frac{1}{EA} \sum N_i^0 \cdot N_i^{II} L_i = \frac{1}{EA} \left[10 \sum N_i^I N_i^{II} L_i + 6 \sum N_i^{II^2} L_i \right]$$

$$\vec{M}_D = \frac{10 \cdot 0 + 6(2 + 4\sqrt{2})}{EA}$$

⊗ Problema 2.1

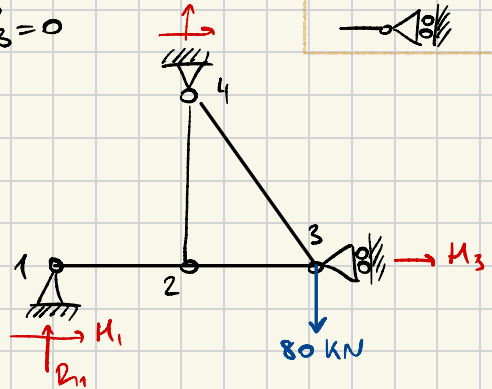
$$EA = 3 \cdot 10^5 \text{ KN}$$



Tenemos SIMETRÍA

$$\vec{U}_3 = 0$$

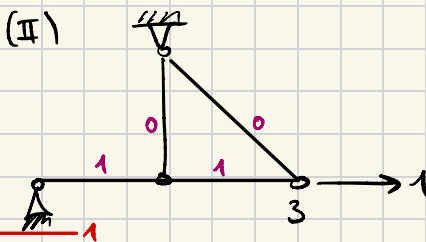
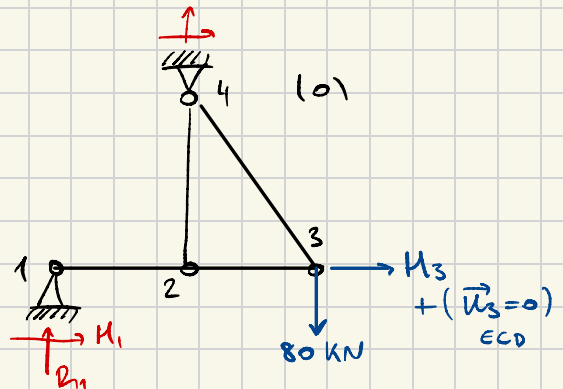
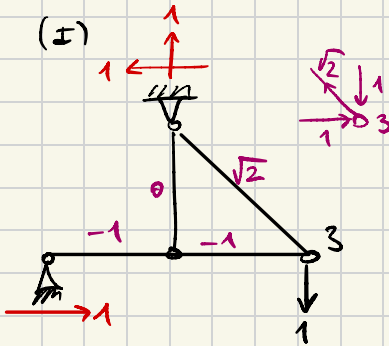
Si hay articula...



$$GH = 5 + 4 - 2 \cdot 4 = 9 - 8 = 1 \text{ (HIPERESTÁTICA)}$$

Si no hubiéramos quitado la barra, tendríamos esto mal.

↳ Liberamos un apoyo



$$N_i^0 = 80 N_i^I + H_3 \cdot N_i^{II}$$

CASTIGLIANO

$$\vec{U}_3 = 0 = \frac{1}{EA} \sum N_i^0 N_i^I L_i = \frac{1}{EA} [80 \sum N_i^I N_i^I L_i + H_3 \sum N_i^I N_i^I L_i] \Rightarrow$$

$$\Rightarrow H_3 = \frac{-80 \sum N_i^F N_i^H L_i}{\sum N_i^H L_i}$$

↓

$$V_3 = \frac{1}{EA} \sum N_i^0 N_i^F L_i = \frac{1}{EA} [80 \sum N_i^{F^2} L_i + H_3 \sum N_i^H N_i^F L_i]$$

B	N_i^F	N_i^H	L_i	$N_i^F N_i^H L_i$	$N_i^{F^2} L_i$	$N_i^{H^2} L_i$	N_i^0
1 2	-1	1	L	-L	L	L	0
2 3	-1	1	L	-L	L	L	0
2 4	0	0	L	0	0	0	0
3 4	$\sqrt{2}$	0	$\sqrt{2}L$	0	$2\sqrt{2}L$	0	$80\sqrt{2}$
				-2L	$(2+2\sqrt{2})L$	2L	

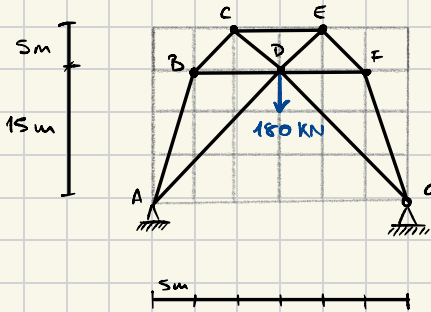
$$H_3 = \frac{-80 \sum N_i^F N_i^H L_i}{\sum N_i^H L_i} = \frac{-80(-2L)}{2L} = 80 \text{ KN}$$

↓

$$V_3 = \frac{1}{EA} \sum N_i^0 N_i^F L_i = \frac{1}{EA} [80 \sum N_i^{F^2} L_i + H_3 \sum N_i^H N_i^F L_i] =$$

$$= \frac{80(2+2\sqrt{2})L + 80(-2L)}{EA} = \frac{160\sqrt{2} \cdot L}{3ES}$$

EXAMEN 17 ENE 2025 (50 min)



$$EA = 10^5 \text{ kN}$$

$$a) GH = LE + b - 2n = 4 + 11 - 2 \cdot 7 = 1$$

Revisar si hay barras que no trabajen.

a) GH

b) Simplificar / simetría

c) Describir célula

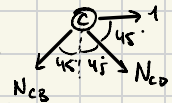
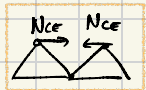
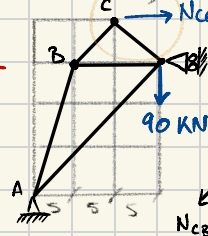
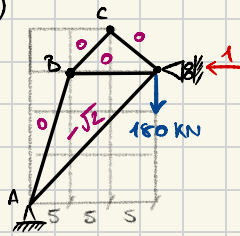
d) Axiles y Reacciones

e) U_D, V_D

$\Rightarrow 0$. Es una estructura simétrica.

Las reacciones en los nudos siempre se paven hacia afuera.

b)



Como es hiperestatica, puedes liberar un apoyo o girar una barra. Si es una estr. simétrica intentar no romper la simetría.

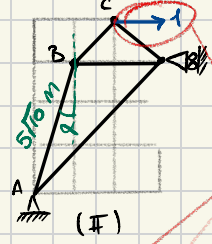
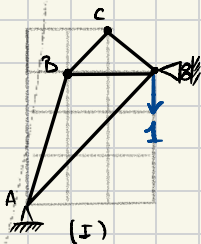
Ecuación de compatibilidad:

$$U_c = \frac{\Delta L_{ce}}{2}$$

$$U_c = \frac{\Delta L_{ce}}{2} = \frac{N_{ce} L_{ce}}{2EA}$$

(SIMETRÍA)

ESQUEMAS AUXILIARES



Iguálamos los sentidos por comodidad.

$$N^{(0)} = 90 \cdot N + N_{ce} N^{II}$$

$$U_c = - \frac{N_{ce} \cdot L_{ce}}{2EA}$$

$$\vec{U}_c = - \frac{N_{ce} L_{ce}}{2EA} \rightarrow \text{Por Castigliano:}$$

$$\frac{1}{EA} \sum N_i^0 \cdot N_i^{II} \cdot L_i = \frac{1}{EA} [90 \sum N_i^I N_i^{II} L_i +$$

$$+ N_{ce} \sum N_i^{II} L_i] = - \frac{N_{ce} L_{ce}}{2EA} \Rightarrow$$

$$N_{ce} = \frac{-90 \sum N_i^I N_i^{II} L_i}{\sum N_i^{II} L_i + \frac{L_{ce}}{2}}$$

$$V_D = \frac{1}{EA} \sum N_i^0 N_i^I L_i = \frac{1}{EA} [90 \sum N_i^{I^2} L_i +$$

$$+ N_{ce} \sum N_i^{II} N_i^I L_i]$$

$$\sum F_V = 0 \rightarrow \frac{N_{CB}}{\sqrt{2}} + \frac{N_{CD}}{\sqrt{2}} = 0$$

$$N_{CB} = -N_{CD}$$

$$\sum F_H = 0 \rightarrow 1 + \frac{N_{CD}}{\sqrt{2}} - \frac{N_{CB}}{\sqrt{2}} = 0$$

$$N_{CD} = -\frac{1}{\sqrt{2}}$$

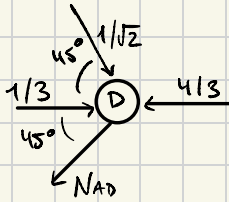
$$\sum F_V = 0 \rightarrow N_{AB} \sin \alpha = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2}$$

$$N_{AB} = \frac{\sqrt{10}}{6}$$

$$\sum F_H = 0 \rightarrow N_{BD} + \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2} = 0$$

$$N_{BD} = -\frac{1}{3}$$

$$1 \cdot 20 = H_D \cdot 15 \rightarrow H_D = \frac{20}{15} = \frac{4}{3}$$



$$\sum F_v = 0 \quad \frac{1}{\sqrt{2}} \frac{\sqrt{2}}{2} + N_{AD} \frac{\sqrt{2}}{2} = 0$$

$$N_{AD} = -1/\sqrt{2}$$

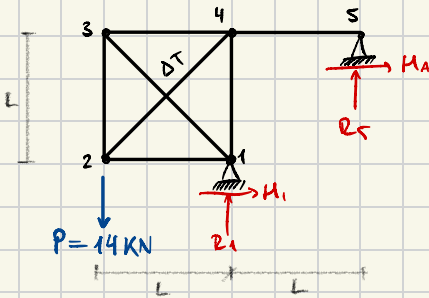
BAHM

	N_i^I	N_i^II	L_i	$N_i^I N_i^{II} L_i$	$N_i^{II^2} L_i$	$N_i^{I^2} L_i$	N^0
AB	0	$\sqrt{10}/6$	$5\sqrt{10}$	0	$50\sqrt{10}/36$	0	-35,7
AD	$-\sqrt{2}$	$-1/\sqrt{2}$	$15\sqrt{2}$	$+15\sqrt{2}$	$15\sqrt{2}/2$	$30\sqrt{2}$	-79,4
BC	0	$1/\sqrt{2}$	$5\sqrt{2}$	0	$5\sqrt{2}/2$	0	-47,9
BD	0	$-1/3$	10	0	$10/9$	0	22,58
CB	0	$-1/\sqrt{2}$	$5\sqrt{2}$	0	$5\sqrt{2}/2$	0	47,9
Σ				$15\sqrt{2}$	23,18	$30\sqrt{2}$	

$$N_{CE} = \frac{-90 \sum N_i^I N_i^{II} L_i}{\sum N_i^{II^2} L_i + \frac{L_{CE}}{2}} = \frac{-90 \cdot (-15\sqrt{2})}{28,18 + 5} = -67,75 \text{ KN}$$

$$\overset{\perp}{V}_D = \frac{1}{EA} [90 \cdot 30\sqrt{2} - 67,75 \cdot 15\sqrt{2}] = \underline{0,0238 \text{ m}}$$

⊗ ORDINARIA ENE24



$EA = 100 \text{ kN}$
 $P = 1 \text{ kN}$
 $\Delta T_{2-4} = 60^\circ \text{C}$
 $\alpha = 10^{-5} \text{ } ^\circ\text{C}^{-1}$

- a) G_H b) Reacciones
 c) N_{4-5} d) $\downarrow V_2$

a) $G_H = LE + b - 2n = 4 + 7 - 2 \cdot 5 = 1.$

b) $R_5 = 0$ $\sum M_1^+ = 0$ $H_5 \cdot L = PL$

$\sum F_H^+ = 0$ $H_5 + H_1 = 0$

$H_5 = P$

$H_1 = -P$

$\sum F_V = 0$ $R_1 - P + R_5 = 0$

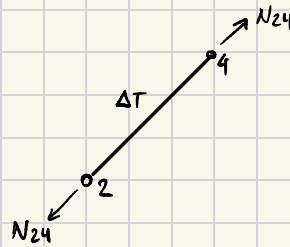
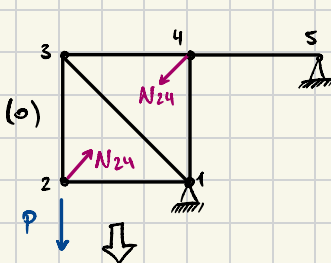
$R_1 = P$

c) $N_{45} = P$

$N_{45} \leftarrow \textcircled{5} \rightarrow H_5$

⊗ Como es hiperestática interna podemos hallar sus reacciones. Tó porque sabemos que habiendo una barra libre en apoyo, una de sus reacciones = 0.

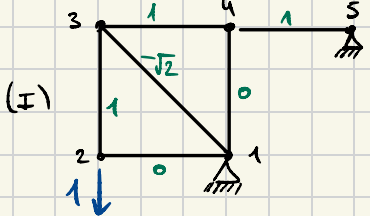
1ª FORMA DE RESOLVERLO : Quito barra 2-4 (La de ΔT)



$\Delta L_{24} = \frac{N_{24} L_{24}}{EA} + \alpha \Delta T L_{24}$

$\rightarrow \leftarrow$
 $\delta_{2-4} = -\Delta L_{2-4}$

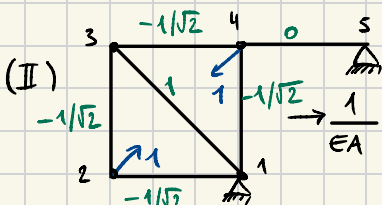
E.C.D (Ec. de Comp. en Deformaciones)



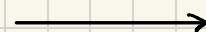
$N_i^{(0)} = P N_i^I + N_{24} \cdot N_i^{II}$

$\frac{1}{EA} \sum N_i^{(0)} N_i^{II} \cdot L_i = \frac{-N_{24} L_{24}}{EA} - \alpha \Delta T L_{24} \rightarrow$

$\rightarrow \leftarrow$
 δ_{2-4}



$\rightarrow \frac{1}{EA} \left[P \sum N_i^I N_i^{II} \cdot L_i + N_{24} \sum N_i^{II} N_i^{II} \cdot L_i \right] = -\frac{N_{24} L_{24}}{EA} - \alpha \Delta T L_{24}$



$$\rightarrow N_{2-4} = \frac{-P \sum N_i^I N_i^{II} L_i - EA \alpha \cdot \Delta T \cdot L_{24}}{\sum N_i^{II^2} L_i + L_{24}} = \frac{-1(-2-\sqrt{2}) - 100 \cdot 10^{-5} \cdot 60\sqrt{2}}{2+\sqrt{2} + \sqrt{2}} =$$

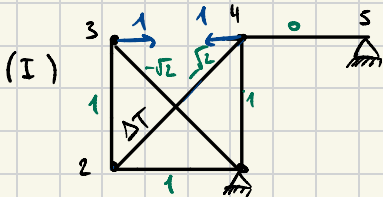
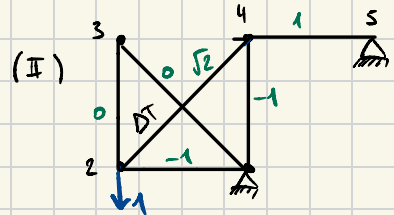
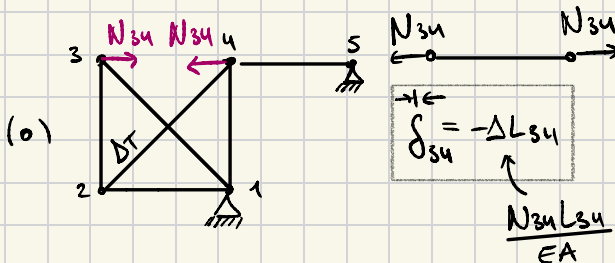
$$N_{2-4} = 0,69 \text{ kN}$$

$$\downarrow \ddot{V}_2 = \frac{1}{EA} \sum N_i^0 \cdot N_i^I L_i = \frac{1}{EA} [P \sum N_i^{I^2} L_i + N_{24} \sum N_i^{II} N_i^I L_i] =$$

$$= \frac{1}{100} [1 \cdot (3+2\sqrt{2}) + 0,69 (-2-\sqrt{2})] \Rightarrow \downarrow \ddot{V}_2 = 0,0347 \text{ m}$$

BARRA	N_i^I	N_i^{II}	L_i	$N_i^I N_i^{II} L_i$	$N_i^{I^2} L_i$	$N_i^{II^2} L_i$	N_i^0
1-2	0	$-1/\sqrt{2}$	1	0	0	1/2	-0,487
1-3	$-\sqrt{2}$	1	$\sqrt{2}$	-2	$2\sqrt{2}$	$\sqrt{2}$	-0,725
1-4	0	$-1/\sqrt{2}$	1	0	0	1/2	-0,487
2-3	1	$-1/\sqrt{2}$	1	$-1/\sqrt{2}$	1	1/2	0,5124
3-4	1	$-1/\sqrt{2}$	1	$-1/\sqrt{2}$	1	1/2	0,5124
4-5	1	0	1	0	1	0	1
				$-2-2\sqrt{2}$	$3+2\sqrt{2}$	$2+\sqrt{2}$	

2ª FORMA DE RESOLVERLO: Quitamos barra 3-4



$$N_i^{(0)} = N_{34} N_i^I + P N_i^II$$

En este caso: $N_{24}^I \cdot \Delta T$ $\alpha \Delta T L_{24}$

$$\frac{1}{EA} \sum N_i^0 N_i^I L_i + \sum N_i^I \cdot \delta_i^{\Delta T} = - \frac{N_{34} \cdot L_{34}}{EA}$$

Variante porque hay temperatura $\rightarrow$ Variante de Casti que estás usando, x las variaciones de t^a .

Puede haber variaciones debido a t^a o a errores de producción.

$$\rightarrow \frac{1}{EA} [N_{34} \sum N_i^{I^2} L_i + P \sum N_i^I N_i^{II} L_i] + N_{24}^I \cdot \alpha \Delta T L_{24} = \frac{N_{34} \cdot L_{34}}{EA} \rightarrow$$

$$\rightarrow N_{34} = \frac{-P \sum N_i^I N_i^{II} L_i - EA N_{24}^I \alpha \Delta T L_{24}}{\sum N_i^{I^2} L_i + L_{34}} = \frac{-1(-2-2\sqrt{2}) - 100(-\sqrt{2}) \cdot 10^{-5} \cdot 60 \cdot \sqrt{2}}{3 + 4\sqrt{2} + 1}$$

$$N_{34} = 0,5124$$

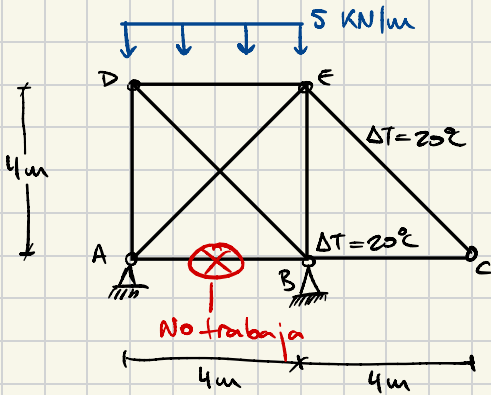
Da lo mismo
OK.

$$\downarrow V_2 = \frac{1}{EA} \sum N_i^0 N_i^{II} L_i + \sum N_i^{II} \cdot \delta_i^{\Delta T} = \frac{1}{100} [N_{34} \sum N_i^I N_i^{II} L_i + P \sum N_i^{II^2} L_i] + \sqrt{2} \cdot 10^{-5} \cdot 60 \cdot \sqrt{2} \rightarrow V_2 = 0,0347 \text{ m}$$

Da lo mismo
OK.

BARRA	N_i^I	N_i^{II}	L_i	$N_i^I N_i^{II} L_i$	$N_i^{I^2} L_i$	$N_i^{II^2} L_i$	N_i^0
1-2	1	-1	1	-1	1	1	
1-3	$-\sqrt{2}$	0	$\sqrt{2}$	0	$2\sqrt{2}$	0	
1-4	1	-1	1	-1	1	1	
2-3	1	0	1	0	1	0	
2-4	$-\sqrt{2}$	$\sqrt{2}$	$\sqrt{2}$	$-2\sqrt{2}$	$2\sqrt{2}$	$2\sqrt{2}$	
4-5	0	1	1	0	0	1	
				$-2-2\sqrt{2}$	$3+4\sqrt{2}$	$3+2\sqrt{2}$	

EX 16 JUN 2025 (45 min)



DATOS

$$EA = 10^5 \text{ KN}$$

$$\alpha = 2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$$

a) Q_H y $P_{roced.}$

b) N_i

c) V_c

d) U_c

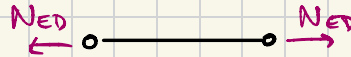
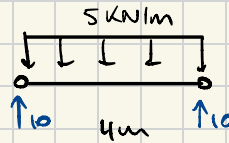
$$a) Q_H = LE + b - 2N = 4 + 8 - 2 \cdot 5 = 10 - 10 = 0$$

$$Q_H = 1$$

Puedes resolverlo de 3 formas: (1) Quitar barra de arriba.

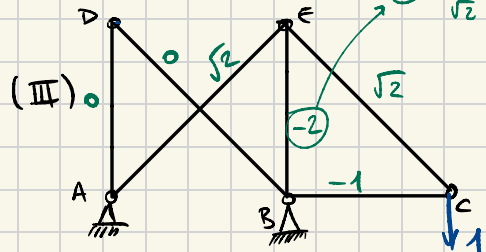
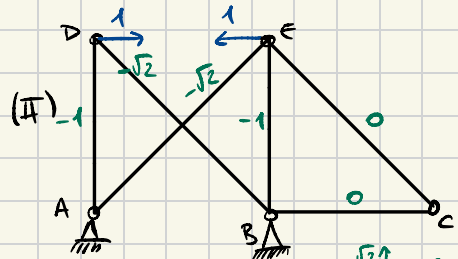
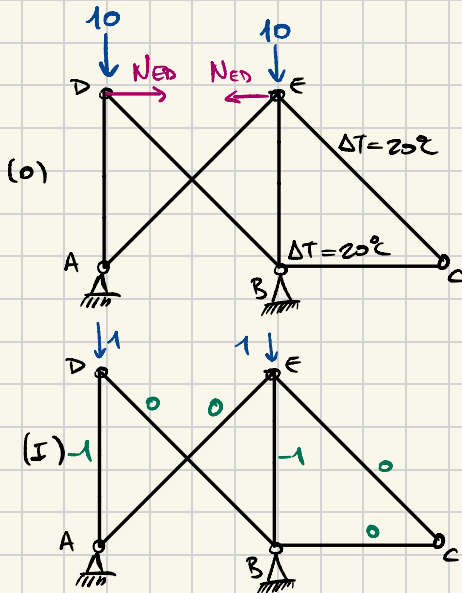
(2) Quitar una cara $T\Delta$. (3) Poner carrito en B.

Elegimos la opción (1)



$\frac{N_{DE} \cdot L_{DE}}{EA}$

$$\delta = -\Delta L_{DE}$$



(II) ΔT Δx $0 \cdot \Delta T + 0 \cdot \Delta T = 0$

$$8 + 16\sqrt{2} + 4$$

(III)

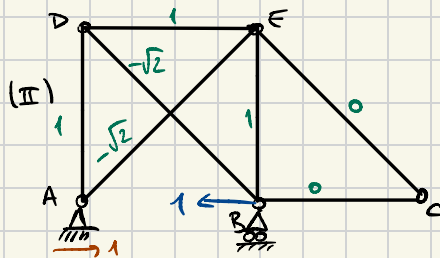
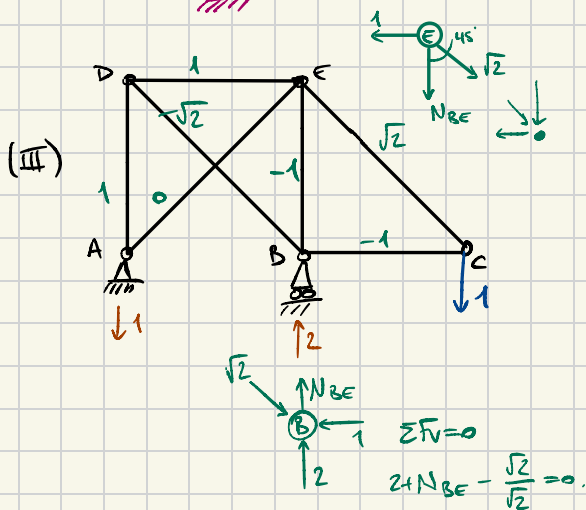
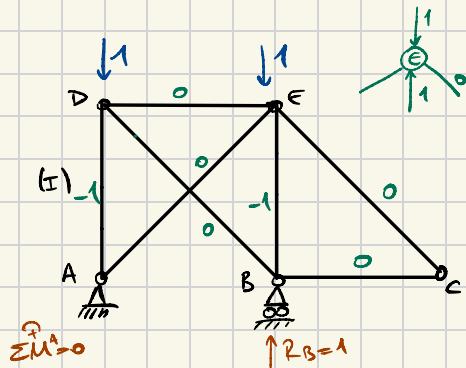
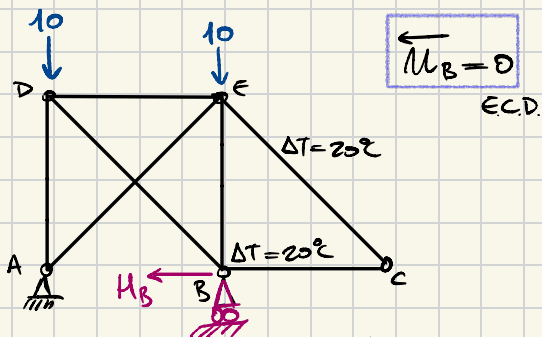
ΔT $\sqrt{2}$

ΔT 1

-8	$8+16\sqrt{2}$	8	$-8-8\sqrt{2}$
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$$\vec{u}_c = \frac{N_{BC} L_{BC}}{EA}$$

Resolvemos con opción 3:



$$\bar{U}_B = \frac{1}{EA} \sum N_i^0 N_i^I L_i + \sum N_i^I \delta_{\Delta T}^I = \frac{1}{EA} \left[10 \sum N_i^I N_i^I L_i + H_B \sum N_i^I L_i \right] = 0$$

$$H_B = \frac{-10 \sum N_i^I N_i^I L_i}{\sum N_i^I L_i} = \frac{-10 (-8)}{12 + 16\sqrt{2}} = 2,31 \text{ kN.}$$

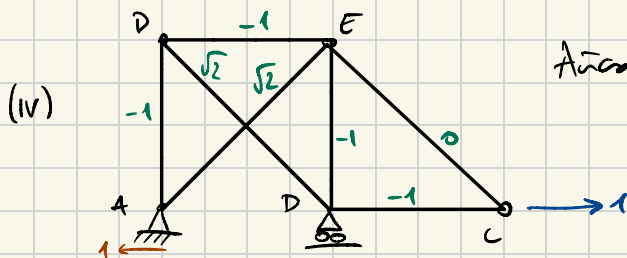
$$\downarrow$$

$$V_c = \frac{1}{EA} \sum N_i^{\circ} N_i^{\text{III}} L_i + \sum N_i^{\text{III}} \delta_i^{\Delta T} = \frac{1}{EA} \left[10 \sum N_i^{\text{I}} N_i^{\text{III}} L_i + \right.$$

$$\left. + H_B \sum N_i^{\text{II}} N_i^{\text{III}} L_i \right] + \overset{2 \cdot 10^{-5}}{\downarrow} \overset{20^{\circ}\text{C}}{\Delta T} \cdot [(-1) \cdot 4 + \sqrt{2} \cdot 4\sqrt{2}] = 10^{-5} [10 \cdot 0 + 2,81 (4 + 8\sqrt{2})] +$$

$$+ 2 \cdot 10^{-5} \cdot 20 \cdot 4 = 195,37 \cdot 10^{-5} \text{ m} = 1,95 \text{ mm}$$

BARRA	N_i^{I}	N_i^{II}	N_i^{III}	L_i	$N_i^{\text{I}} N_i^{\text{III}} L_i$	$N_i^{\text{II}^2} L_i$	$N_i^{\text{I}} N_i^{\text{III}} L_i$	$N_i^{\text{II}} N_i^{\text{III}} L_i$	N_i°
AD	-1	1	1	4	-4	4	-4	4	-7,69
AE	0	$-\sqrt{2}$	0	$4\sqrt{2}$	0	$8\sqrt{2}$	0	0	-3,24
BD	0	$-\sqrt{2}$	$-\sqrt{2}$	$4\sqrt{2}$	0	$8\sqrt{2}$	0	$8\sqrt{2}$	-3,27
BE	-1	1	-1	4	-4	4	4	-4	-7,69
BC	0	0	-1	4	0	0	0	0	0
EC	0	0	$\sqrt{2}$	$4\sqrt{2}$	0	0	0	0	0
DE	0	1	1	4	0	4	0	4	2,81
					-8	$16 + 16\sqrt{2}$	0	$4 + 8\sqrt{2}$	



Añadimos

$N_i^{\text{II}} N_i^{\text{IV}} L_i$

$$\vec{u}_c = \frac{1}{EA} \sum N_i^{\circ} N_i^{\text{IV}} L_i + \sum N_i^{\text{IV}} \delta_i^{\Delta T} = \frac{1}{EA} \left[10 \sum N_i^{\text{I}} N_i^{\text{IV}} L_i + H_B \sum N_i^{\text{II}} N_i^{\text{IV}} L_i \right] + \alpha \Delta T (\dots)$$

$$= \frac{1}{10^5} [10 \cdot 8 + 2,81 (-12 - 16\sqrt{2})] + 2 \cdot 10^{-5} \cdot 20 \cdot (1 \cdot 4) = 1605 \cdot 10^{-5} \text{ m} = 1,6 \text{ mm}$$

BARRA	N_i^{I}	N_i^{II}	N_i^{III}	L_i	N_i^{IV}	$N_i^{\text{I}} N_i^{\text{IV}} L_i$	$N_i^{\text{II}} N_i^{\text{IV}} L_i$
AD	-1	1	1	4	-1	4	-4
AE	0	$-\sqrt{2}$	0	$4\sqrt{2}$	$\sqrt{2}$	0	$-8\sqrt{2}$
BD	0	$-\sqrt{2}$	$-\sqrt{2}$	$4\sqrt{2}$	$\sqrt{2}$	0	$-8\sqrt{2}$
BE	-1	1	-1	4	-1	4	-4
BC	0	0	-1	4	1	0	0
EC	0	0	$\sqrt{2}$	$4\sqrt{2}$	0	0	0
DE	0	1	1	4	-1	0	-4
					8	-12	$-16\sqrt{2}$